



Exponential finite-difference method applied to Korteweg–de Vries equation for small times

A. Refik Bahadır

Department of Mathematics, Faculty of Arts and Sciences, İnönü University, 44100 Malatya, Turkey

Abstract

The Korteweg–de Vries equation is numerically solved by using the exponential finite-difference technique. The accuracy of computed solutions is examined by comparison with other numerical and analytical solutions using two examples. The close results agreement between the current results and the exact solutions confirms that the proposed finite-difference procedure is an effective technique for the solution of the Korteweg–de Vries equation at the small times.

© 2003 Elsevier Inc. All rights reserved.

Keywords: Korteweg–de Vries equation; Exponential finite-differences

1. Introduction

A considerable interest in studying the Korteweg–de Vries equation (referred to as the KdV equation henceforth) by physicists and mathematicians has been generated since the discovery of solutions by Zabusky and Kruskal [1]. Several methods of solving this equation both analytically and numerically have been proposed. Analytic solutions to particular problems have been produced by [2,3] and numerical solutions based on finite-difference techniques [4,5], pseudospectral methods [3], heat balance integral method [6] and finite element techniques [7–11]. A survey of results and applications for the KdV equation was written by Miura [12].

E-mail address: arbahadir@inonu.edu.tr (A. Refik Bahadır).

In the present work we apply the exponential finite-difference technique to solve the KdV equation.

2. The governing equation and numerical method

If the KdV equation is taken in the form

$$\frac{\partial U}{\partial t} + \varepsilon U \frac{\partial U}{\partial x} + \mu \frac{\partial^3 U}{\partial x^3} = 0, \quad a \leq x \leq b, \quad t > 0, \quad (1)$$

where ε and μ are positive parameters. We will now compute solutions to Eq. (1) for the following problems.

Problem 1. As a model of single soliton we first take as initial condition

$$U(x, 0) = 3C \operatorname{sech}^2(Ax + D), \quad 0 \leq x \leq 2$$

with the boundary conditions

$$U(0, t) = U(2, t) = 0, \quad t > 0.$$

The analytic solution to this problem can be expressed as [13]:

$$U(x, t) = 3C \operatorname{sech}^2(Ax - Bt + D), \quad 0 \leq x \leq 2,$$

where C and D are constants, $A = \frac{1}{2\sqrt{\frac{\varepsilon C}{\mu}}}$ and $B = \varepsilon AC$.

Problem 2. As the second problem we shall study will be two-solitons solution subject to the conditions; boundary conditions

$$U(0, t) = U(4, 0) = 0, \quad t > 0$$

and the initial conditions shall be derived from the exact solution [14]

$$U(x, t) = 12\mu(\log E)_{xx}, \quad 0 \leq x \leq 4,$$

where

$$E = 1 + \exp(\eta_1) + \exp(\eta_2) + \left(\frac{\alpha_1 - \alpha_2}{\alpha_1 + \alpha_2} \right)^2 \exp(\eta_1 + \eta_2),$$

$$\eta_i = \alpha_i x - \alpha_i^3 \mu t + b_i, \quad (i = 1, 2),$$

$$b_1 = -0.48\alpha_1, \quad b_2 = -1.07\alpha_2, \quad \alpha_1 = \sqrt{\frac{0.3}{\mu}}, \quad \alpha_2 = \sqrt{\frac{0.1}{\mu}}.$$

The exponential finite-difference method that we applied to solve Problems 1 and 2 was originally developed by Bhattacharya [15] and used to solve one-

dimensional heat conduction in a solid slab [16,17]. It is also used to model a heat conduction problem in three-dimensions [18].

We assume that $F(U)$ denote any continuous differential function. Multiplying Eq. (1) by the derivative of F leads to the following equation:

$$\frac{\partial F}{\partial U} \frac{\partial U}{\partial t} = F'(U) \left(-\varepsilon U \frac{\partial U}{\partial x} - \mu \frac{\partial^3 U}{\partial x^3} \right)$$

and

$$\frac{\partial F}{\partial t} = F'(U) \left(-\varepsilon U \frac{\partial U}{\partial x} - \mu \frac{\partial^3 U}{\partial x^3} \right). \quad (2)$$

Using the usual forward difference replacement for $\frac{\partial F}{\partial t}$ we obtain the finite-difference representation of Eq. (2) as

$$F(U_i^{n+1}) = F(U_i^n) + kF'(U_i^n) \left[-\varepsilon U_i^n \left(\frac{\partial U}{\partial x} \right)_i^n - \mu \left(\frac{\partial^3 U}{\partial x^3} \right)_i^n \right],$$

where k is the time step. Now, if we let $F(U) = \ln U$ then we obtained the exponential finite-difference scheme as

$$U_i^{n+1} = U_i^n + \exp \left\{ \frac{k}{U_i^n} \left[-\varepsilon U_i^n \left(\frac{\partial U}{\partial x} \right)_i^n - \mu \left(\frac{\partial^3 U}{\partial x^3} \right)_i^n \right] \right\}.$$

The finite-differences for the derivatives have been taken following form:

$$\left(\frac{\partial U}{\partial x} \right)_i^n = \frac{-3U_i^n + 4U_{i+1}^n - U_{i+2}^n}{2h} \quad (3)$$

and

$$\left(\frac{\partial^3 U}{\partial x^3} \right)_i^n = \frac{-5U_i^n + 18U_{i+1}^n - 24U_{i+2}^n + 14U_{i+3}^n - 3U_{i+4}^n}{2h^3}, \quad i = 1, \quad (4)$$

$$\left(\frac{\partial U}{\partial x} \right)_i^n = \frac{U_{i+1}^n - U_{i-1}^n}{2h} \quad (5)$$

and

$$\left(\frac{\partial^3 U}{\partial x^3} \right)_i^n = \frac{U_{i+2}^n - 2U_{i+1}^n + 2U_{i-1}^n - U_{i-2}^n}{2h^3}, \quad i = 2, \quad N - 2, \quad (6)$$

$$\left(\frac{\partial U}{\partial x} \right)_i^n = \frac{3U_i^n - 4U_{i-1}^n + U_{i-2}^n}{2h} \quad (7)$$

and

$$\left(\frac{\partial^3 U}{\partial x^3}\right)_i^n = \frac{5U_i^n - 18U_{i-1}^n + 24U_{i-2}^n - 14U_{i-3}^n + 3U_{i-4}^n}{2h^3}, \quad i = N - 1. \quad (8)$$

Numerical solutions obtained by the above scheme is compared with the solutions obtained from other two numerical schemes, classical explicit finite-difference (EFD) and heat balance integral (HBI) method as well as the exact solutions. The classical explicit finite-difference equations can be obtained from (3)–(8).

The HBI solutions are given as [6]

$$U(x, t) = 3C \exp(3\mu t/2) \operatorname{sech}^2(Ax + D), \quad 0 \leq x \leq 2, \quad t \geq 0$$

and

$$U(x, t) = 12\mu \exp(3\mu t/16) (\log E)_{xx}, \quad 0 \leq x \leq 4, \quad t \geq 0$$

for the Problems 1 and 2 respectively.

3. Numerical results and discussion

All numerical computations we performed with the space step $h = 0.1$ and the time step $k = 0.0001$. The Problems 1 and 2 respectively concern the generation of a single solitary wave and the interaction of two well separated

Table 1
Comparison numerical and exact solutions of the Problem 1 at $t = 0.005$

x	Exponential	EFD	HBI	Exact
0.0	0.0	0.0	0.0	0.0
0.1	0.00224711	0.00026665	0.00026665	0.00025688
0.2	0.00355567	0.00320906	0.00320978	0.00309232
0.3	0.03777912	0.03794673	0.03794787	0.03658529
0.4	0.35698115	0.36617631	0.36617907	0.35574607
0.5	0.86070242	0.85627247	0.85626815	0.86305608
0.6	0.17729622	0.17201488	0.17201146	0.17787428
0.7	0.01512775	0.01568136	0.01568004	0.01627120
0.8	0.00113721	0.00131132	0.00131099	0.00136083
0.9	0.00009312	0.00010884	0.00010880	0.00011294
1.0	0.00000779	0.00000903	0.00000902	0.00000937
1.1	0.00000065	0.00000075	0.00000075	0.00000078
1.2	0.00000005	0.00000006	0.00000006	0.00000006
1.3	0.0	0.00000001	0.00000001	0.00000001
1.4	0.0	0.0	0.0	0.0
⋮	⋮	⋮	⋮	⋮
2.0	0.0	0.0	0.0	0.0

Table 2

Comparison numerical and exact solutions of the Problem 1 at $t = 0.01$

x	Exponential	EFD	HBI	Exact
0.0	0.0	0.0	0.0	0.0
0.1	0.00405629	0.00026665	0.00026665	0.00024746
0.2	0.00389535	0.00320906	0.00320979	0.00297915
0.3	0.03764657	0.03794673	0.03794800	0.03527076
0.4	0.34792815	0.36616631	0.36618040	0.34551632
0.5	0.86481172	0.85627247	0.85627126	0.86931951
0.6	0.18273885	0.17201488	0.17201208	0.18391225
0.7	0.01458343	0.01568136	0.01568009	0.01688449
0.8	0.00095558	0.00131132	0.00131099	0.00141257
0.9	0.00007766	0.00010884	0.00010881	0.00011724
1.0	0.00000672	0.00000903	0.00000902	0.00000972
1.1	0.00000056	0.00000075	0.00000075	0.00000081
1.2	0.00000005	0.00000006	0.00000006	0.00000007
1.3	0.0	0.00000001	0.00000001	0.00000001
1.4	0.0	0.0	0.0	0.0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
2.0	0.0	0.0	0.0	0.0

solitons, which are accomplished by taking the parameters $\varepsilon = 1.0$, $\mu = 4.84 \times 10^{-4}$, $C = 0.3$ and $D = -6.0$. The exponential finite-difference solutions are compared with the EFD, HBI and the exact solution at the nodes. The results obtained for both problems are displayed in Tables 1–4 for times $t = 0.005$ and 0.01 . According to the results presented here, the exponential

Table 3

Comparison numerical and exact solutions of the Problem 2 at $t = 0.005$

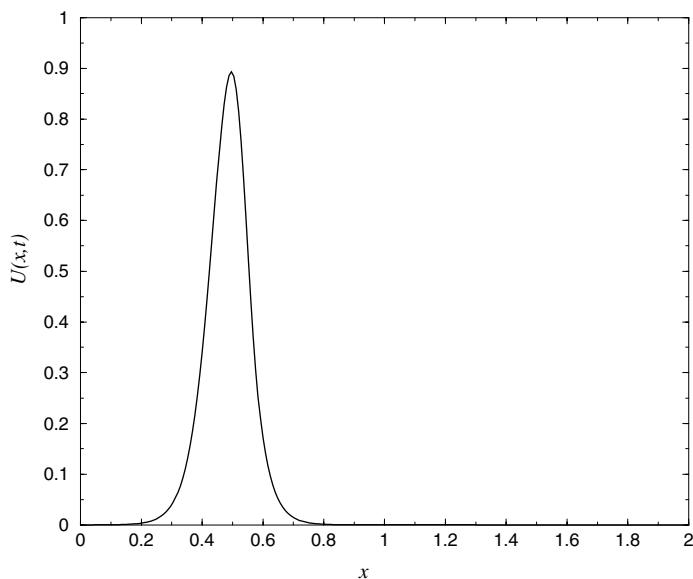
x	Exponential	EFD	HBI	Exact
0.0	0.0	0.0	0.0	0.0
0.2	0.00161957	0.00130445	0.00146231	0.00140886
0.4	0.16250630	0.16402277	0.16469787	0.16006740
0.6	0.07258729	0.07123047	0.07126321	0.07370329
0.8	0.00124599	0.00137135	0.00130927	0.00132438
1.0	0.01296958	0.01290987	0.01295362	0.01286574
1.2	0.11226482	0.11251313	0.11258566	0.11228939
1.4	0.05036881	0.05013744	0.05012470	0.05040753
1.6	0.00348081	0.00353736	0.00350801	0.00353296
1.8	0.00019821	0.00020283	0.00020051	0.00020196
2.0	0.00001119	0.00001146	0.00001132	0.00001140
2.2	0.00000063	0.00000065	0.00000064	0.00000064
2.4	0.00000004	0.00000004	0.00000004	0.00000004
2.6	0.00000001	0.00000001	0.00000001	0.00000001
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
4.0	0.0	0.0	0.0	0.0

Table 4

Comparison numerical and exact solutions of the Problem 2 at $t = 0.01$

x	Exponential	EFD	HBI	Exact
0.0	0.0	0.0	0.0	0.0
0.2	0.00177650	0.00114718	0.00146231	0.00135737
0.4	0.16032857	0.16335221	0.16469794	0.15552267
0.6	0.07393146	0.07119635	0.07126324	0.07621798
0.8	0.00118044	0.00143272	0.00130927	0.00134032
1.0	0.01298586	0.01286635	0.01295362	0.01277843
1.2	0.11194221	0.11244069	0.11258571	0.11199146
1.4	0.05061504	0.05015009	0.05012472	0.05069147
1.6	0.00345329	0.00356666	0.00350801	0.00355809
1.8	0.00019593	0.00020517	0.00020051	0.00020341
2.0	0.00001106	0.00001159	0.00001132	0.00001149
2.2	0.00000062	0.00000065	0.00000064	0.00000065
2.4	0.00000004	0.00000004	0.00000004	0.00000004
2.6	0.0	0.0	0.0	0.0
⋮	⋮	⋮	⋮	⋮
4.0	0.0	0.0	0.0	0.0

finite-difference scheme behaves better than the other two numerical schemes at small times. Figs. 1 and 2 show that the exponential finite-difference solutions also satisfy the physical behavior of the problems.

Fig. 1. Exponential finite-difference solution of the Problem 1 at $t = 0.01$.

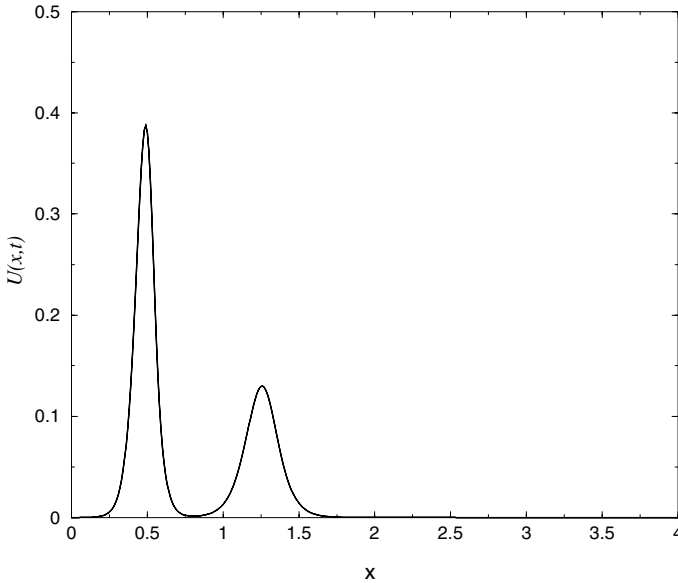


Fig. 2. Exponential finite-difference solution of the Problem 2 at $t = 0.01$.

4. Conclusion

The exponential finite-differences method is applied to solve the KdV equation. The technique exhibits higher accuracy than the other numerical methods with which it is compared. Therefore, it is concluded that the exponential finite-differences method can be used to produce reasonably accurate numerical solutions of the KdV equations at small times.

Acknowledgements

This work was supported by grant no. 2002/14 of the İnönü University Research Fund (İUBAP).

References

- [1] N.J. Zabusky, M.D. Kruskal, Interaction of solitons in a collisionless plasma and the recurrence of initial states, *Phys. Rev. Lett.* 15 (1965) 240–243.
- [2] N.J. Zabusky, A synergetic approach to problem of nonlinear dispersive wave propagation and interaction, in: W. Ames (Ed.), *Proc. Symp. Nonlinear Partial Differential Equations*, Academic Press, New York, 1967, pp. 223–258.

- [3] B. Fornberg, G.B. Whitham, A numerical and theoretical study of certain nonlinear wave phenomena, *Philos. Trans. Roy. Soc.* 289 (1978) 373–404.
- [4] K. Goda, On stability of some finite difference schemes for the Korteweg–De Vries equation, *J. Phys. Soc. Japan* 39 (1975) 229–236.
- [5] A.C. Vliengenthart, On finite difference methods for the Korteweg–De Vries equation, *J. Eng. Math.* 5 (1971) 137–155.
- [6] A.R. Bahadır, S. Kutluay, A. Özdeş, A small time solutions for the Korteweg–De Vries equation, *Appl. Math. Comput.* 107 (2000) 203–210.
- [7] I.S. Greig, J.L. Morris, A hopscotch method for the Korteweg–De Vries equation, *J. Comput. Phys.* 20 (1976) 64–80.
- [8] S.W. Schoombie, Spline Petrov–Galerkin methods for the numerical solution of the Korteweg–De Vries equation, *IMA J. Numer. Anal.* 2 (1982) 95–109.
- [9] J.M. Sanz Serna, I. Christie, Petrov Galerkin methods for nonlinear dispersive waves, *J. Comput. Phys.* 39 (1981) 94–102.
- [10] J.L. Bona, V.A. Douglas, O.A. Karakashian, Fully discrete Galerkin methods for the Korteweg–De Vries equation, *Comput. Math. Appl.* 12A (1986) 859–884.
- [11] G.A. Gardner, A.H.A. Ali, L.R.T. Gardner, A finite element solution for the Korteweg–De Vries equation using cubic B-spline shape functions, in: R. Gruber, J. Periaux, R.P. Shaw (Eds.), *ISNME-89*, vol. 2, Springer, Berlin, 1989, pp. 565–570.
- [12] R.M. Miura, The Korteweg–de Vries equation: a survey of results, *SIAM Rev.* 18 (1976) 412–459.
- [13] M.E. Alexander, J.L. Morris, Galerkin methods for some model equations for nonlinear dispersive waves, *J. Comput. Phys.* 30 (1979) 428–451.
- [14] T.R. Taha, M.J. Ablowitz, Analytic and numerical aspects of certain non-linear evolution equations III. Numerical KdV equation, *J. Comp. Phys.* 55 (1984) 231–253.
- [15] M.C. Bhattacharya, An explicit conditionally stable finite difference equation for heat conduction problems, *Int. J. Numer. Meth. Eng.* 21 (2) (1985) 239–265.
- [16] M.C. Bhattacharya, A new improved finite difference equation for heat transfer during transient change, *Appl. Math. Model.* 10 (1) (1986) 68–70.
- [17] M.C. Bhattacharya, M.G. Davies, The comparative performance of some finite difference equations for transient heat conduction problems, *Int. J. Numer. Meth. Eng.* 21 (7) (1987) 1317–1331.
- [18] R.F. Handschuh, T.G. Keith Jr., Applications of an exponential finite-difference technique, *Numer. Heat Transfer Part A* 22 (1992) 363–378.